

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2024

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Expand and simplify $6(x + 2) - 2x$. [2]

2. A school jersey costs \$80. The price rises by 25%. Find the new price. [3]

3. Express 36 as a product of its prime factors. [3]

4. In a class of Moleli, $n(A) = 11$ (play soccer), $n(A \cap B) = 3$ (play both). How many play soccer only? [2]

5. Find the HCF and LCM of 12 and 42. [3]

6. Work out $\frac{2}{3} + \frac{1}{5}$. Give a mixed number if possible. [4]

7. A map of Harare uses scale 3 : 700000. A road is 15 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm $\leftrightarrow$ 1 km if 3:7 is first simplified — instead: scale 3 cm to 7 km.) [4]

8. Write 30000 in standard form. [3]

- 9.** Overnight in Harare the temperature was 5°C . By 14:00 it was 23°C . Find the rise. **[3]**
- 10.** Simplify $2^2 \times 2^2$. Leave as a power of 2. **[3]**
- 11.** In triangle ABC, $\hat{A} = 64^{\circ}$ and $\hat{B} = 45^{\circ}$. Find $\hat{C}$. **[3]**
- 12.** A cylindrical drum has radius 7 cm and height 4 cm. Find its volume. $\pi = 22/7$. **[4]**
- 13.** Factorise $x^2 + 3x + 2$. **[4]**
- 14.** Solve
$$\begin{cases} x + y = 6 \\ x - y = 0 \end{cases}$$
 [4]
- 15.** Factorise completely $6x + 15$. **[3]**
- 16.** A bag contains 5 red, 4 blue and 1 green beads. $P(\text{red})$? **[3]**
- 17.** Find the midpoint of (4, 1) and (6, 7). **[3]**
- 18.** Simple interest on \$800 at 10% per year for 2 years. Find the interest. **[4]**
- 19.** A right-angled triangle has shorter sides 9 cm and 12 cm. Find the hypotenuse. **[4]**
- 20.** The n th term is $7 + 3(n - 1)$. Find the 8th term. **[3]**

- 21.** In right-angled triangle PQR, $\hat{P} = 30^\circ$ and hypotenuse PR = 12 cm. Find PQ, opposite 30° . **[4]**
- 22.** Circumference of a circle radius 21 cm. Take $\pi = 22/7$. **[4]**
- 23.** Make h the subject of $A = 2\pi r(r + h)$. **[4]**
- 24.** Find the mean of 9, 6, 7, 5, 5. **[3]**
- 25.** A regular polygon has 10 sides. Find each interior angle. **[4]**
- 26.** Solve $6x + 4 = 34$. **[3]**
- 27.** Find the gradient of the line joining (5, 6) and (9, 10). **[3]**
- 28.** Work out $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 5 \\ 4 \end{pmatrix}$. **[4]**
- 29.** A triangular plot has base 12 m and perpendicular height 9 m. Find its area. **[3]**
- 30.** Solve $2x + 2 < 10$. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).